
Algorithm 4 Skip Module Hyperparameter Selection

Require: Parameter space Θ , validation set D_{val} , DL model $\mathcal{M}$, recall tolerance δ_R , weights w_S, w_R

Ensure: Optimal parameters θ^*

1: baseline $\leftarrow$ EVALUATEPIPELINE(D_{val} , skip=None, $\mathcal{M}$)

2: Extract R_{base} from baseline

3: $\Theta_{\text{feasible}} \leftarrow \emptyset$

▷ **Pass 1: Collect feasible candidates**

4: **for** each $\theta \in \Theta$ **do**

5: results $\leftarrow$ EVALUATEPIPELINE(D_{val} , $\mathcal{S}(\theta)$, $\mathcal{M}$)

6: Extract $R(\theta)$, $S_r(\theta)$ from results

7: **if** $R(\theta) \geq R_{\text{base}} - \delta_R$ **then**

8: $\rho(\theta) \leftarrow 1 - \frac{R_{\text{base}} - R(\theta)}{\delta_R}$

9: $\Theta_{\text{feasible}} \leftarrow \Theta_{\text{feasible}} \cup \{(\theta, \rho(\theta), S_r(\theta))\}$

10: **end if**

11: **end for**

▷ **Pass 2: Range-normalize and select**

12: $\tilde{\rho}(\theta) \leftarrow \frac{\rho(\theta) - \min_{\Theta_{\text{feasible}}} \rho}{\max_{\Theta_{\text{feasible}}} \rho - \min_{\Theta_{\text{feasible}}} \rho}$ for each $\theta \in \Theta_{\text{feasible}}$

13: $\tilde{S}_r(\theta) \leftarrow \frac{S_r(\theta) - \min_{\Theta_{\text{feasible}}} S_r}{\max_{\Theta_{\text{feasible}}} S_r - \min_{\Theta_{\text{feasible}}} S_r}$ for each $\theta \in \Theta_{\text{feasible}}$

14: $\theta^* \leftarrow \arg \max_{\theta \in \Theta_{\text{feasible}}} [w_R \tilde{\rho}(\theta) + w_S \tilde{S}_r(\theta)]$

15: **return** θ^*
